

Tutorial series 4

Exercise 1: Give the domain of definition of f :

$$f(x) = \frac{x^3+3x+2}{x^2-2x-3} ; f(x) = \exp(\sqrt{1+\sin(x)}) ; f(x) = \sqrt{\sqrt{x+2}-\sqrt{x-2}} + \frac{1}{\sqrt{1-x}}$$

$$f(x) = \sqrt{\frac{2+3x}{5-2x}} ; f(x) = \left(\frac{1+x}{1-x}\right)^{\frac{1}{\sin(x)}} ; f(x) = \ln\left(\frac{x^2-3x+2}{x+1}\right)$$

Exercise 2: Calculate the limit of these functions:

$$\lim_{x \rightarrow -\infty} (\sqrt{x^2+x+1} - \sqrt{x^2+ax+b}) , a, b \in \mathbb{R} ; \lim_{x \rightarrow 1} \left(\frac{x^n-1}{x^p-1}\right) , n, p \in \mathbb{N}^* ; \lim_{x \rightarrow \pi} \left(\frac{\sin(x)}{1-\frac{x}{\pi}}\right)$$

$$\lim_{x \rightarrow +\infty} \sqrt{x + \sqrt{x + \sqrt{x}}} - \sqrt{x} ; \lim_{x \rightarrow 0} \left(\frac{\cos(ax) - \cos(bx)}{x^2}\right) , a, b \in \mathbb{Z}^* ; \lim_{x \rightarrow +\infty} \sqrt{x+5} - \sqrt{x-3}$$

$$\lim_{x \rightarrow 2} \frac{x^2-4}{x^2-3x+2} ; \lim_{x \rightarrow 0} \frac{\sqrt{1+x}-\sqrt{1+x^2}}{x} ; \lim_{x \rightarrow 0} \frac{x^2+2|x|}{x} ; \lim_{x \rightarrow -\infty} \frac{x^2+2|x|}{x}$$

Exercise 3:

1. Study the continuity of these functions:

$$f(x) = \begin{cases} \sin(\pi x) & 0 < x < 1 \\ \ln(x) & 1 \leq x < 2 \end{cases} ; g(x) = \begin{cases} \frac{\cos(x)-\sin(x)}{x-\frac{\pi}{4}} & x < \frac{\pi}{4} \\ -\sqrt{2} + \frac{\pi}{4} - x & x \geq \frac{\pi}{4} \end{cases} ; h(x) = \begin{cases} xE\left(\frac{1}{x}\right) & x \neq 0 \\ 1 & x = 0 \end{cases}$$

2. Do these functions admit an extension by continuity at the point x_0 .

$$f(x) = \frac{\sin(x-5)}{\sqrt{x+4}-3}, x_0 = 5 ; f(x) = x^2 \left[4 - x^2 E\left(\frac{1}{x}\right)\right], x_0 = 0 ; f(x) = \frac{\sin^2(\pi x)}{x-1}, x_0 = 1$$

$$f(x) = \frac{1}{1-x} - \frac{2}{1-x^2}, x_0 = \pm 1 ; f(x) = \sin(x) \sin\left(\frac{1}{x}\right), x_0 = 0 ;$$

Exercise 4:

1. Prove that the function $f(x) = |x^2 - 1|$ is not differentiable at the point 1, but differentiable on the left and on the right of 1.

2. Study the differentiability of these functions:

$$f(x) = \begin{cases} \sin(x) & |x| \geq 0 \\ x & |x| < 0 \end{cases} ; g(x) = \begin{cases} x^2 e^{-x^2} & |x| \leq 1 \\ \frac{1}{e} & |x| > 1 \end{cases}$$

3. Determine the values of a, b, c, d so that the function f will be differentiable in $\mathbb{R}$

$$f(x) = \begin{cases} 4x & x \leq 0 \\ ax^2 + bx + c & 0 < x < 1 \\ x^3 - 2x & x \geq 1 \end{cases} ; f(x) = \begin{cases} ax + b & x \leq 0 \\ cx^2 + dx & 0 < x \leq 1 \\ 1 - \frac{1}{x} & x > 1 \end{cases}$$

Exercise 5:

1. Using the mean value theorem, prove that:

$$\begin{aligned}\forall x > 0, \quad & x + 1 < e^x < xe^x + 1 \\ \forall x, y > 0, \quad & \frac{x-y}{x} < \ln\left(\frac{x}{y}\right) < \frac{x-y}{y} \\ \forall x > 0, \quad & \frac{x-1}{2\sqrt{x}} < \sqrt{x} - 1 < \frac{x-1}{2}\end{aligned}$$

2. Deduce these limits:

$$\lim_{x \rightarrow 1^+} \frac{\sqrt{x}-1}{x-1} \quad ; \quad \lim_{x \rightarrow 1^+} (\sqrt{x})^{\frac{1}{x-1}}$$

Exercise 6: Using the Hospital's rule, calculate:

$$\lim_{x \rightarrow 0} \frac{x \cos(x) - \sin(x)}{x^3} \quad ; \quad \lim_{x \rightarrow 1} \frac{1-x}{1-\sin\left(\frac{\pi x}{2}\right)} \quad ; \quad \lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\ln(x)} \right) \quad ; \quad \lim_{x \rightarrow +\infty} \frac{e^x + e^{-x}}{e^x - e^{-x}} \quad ; \quad \lim_{x \rightarrow +\infty} \frac{x}{e^{ax}}$$

Exercise 7:

1. Calculate the derivate of order n of:

$$f(x) = \frac{x+1}{1-x} \quad ; \quad g(x) = e^x \sin(x) \quad ; \quad h(x) = e^x \cos(x)$$

2. Using the Leibniz rule, calculate $f^{(3)}$ such that: $f(x) = (1+x^2)e^{-2x}$

Exercise 8:

1. Write the Taylor formula at the point a of the function f and give the formula of $\cos(x)$ and $\sin(x)$.
2. Let $f(x) = \ln(x+1)$
 - a. Calculate its derivative of order n .
 - b. Write the Taylor formula of f .
 - c. Calculate $\lim_{x \rightarrow +\infty} \left(1 - \frac{1}{2} + \frac{1}{3} + \dots + \frac{(-1)^{n+1}}{n} \right)$
 - d. Using Taylor formula of order 2 then of order 1, prove that:

$$\forall x > 0, x - \frac{x^2}{2} < \ln(x+1) < x$$

Exercise 9:

Give the limited development of order n , in the neighborhood of x_0 of the following functions:

1. $f(x) = \frac{e^x}{(1+x)^3} \quad n = 2, \quad x_0 = 0$
2. $f(x) = \frac{\sin x}{1+\ln(1+x)} \quad n = 3, \quad x_0 = 0$
3. $f(x) = \cos(\pi \sin(x)) \quad n = 4, \quad x_0 = \frac{\pi}{2}$
4. $f(x) = x^{\ln(x)} \quad n = 4, \quad x_0 = 1$
5. $f(x) = \sqrt{\frac{1+x}{2+x}} \quad n = 3, \quad x_0 \rightarrow +\infty$

6. $f(x) = (1 + \frac{1}{x})^x$ $n = 2$, $x_0 \rightarrow +\infty$

Exercise 10: Calculate these limits using the limited development.

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)-x}{e^x - \sqrt{1+2x}} \quad ; \quad \lim_{x \rightarrow e} \frac{\sqrt{x}-\sqrt{e}}{\ln(x)-1} \quad ; \quad \lim_{x \rightarrow +\infty} (x \sin(\frac{1}{x}))^{x^2}$$

Exercise 11: Determine the domain of definition, then calculate the derivative of:

$$f(x) = \sin^{-1}\left(\frac{x}{x+1}\right) \quad ; \quad f(x) = \cos^{-1}(\sqrt{2-x^2}) \quad ; \quad f(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

$$f(x) = \cos^{-1}(\ln(x)) \quad ; \quad f(x) = \sin^{-1}(1-x) \quad ; \quad f(x) = \tan^{-1}(1-\ln(x))$$

$$f(x) = \sinh^{-1}(\sqrt{x^2-1}) \quad ; \quad f(x) = \cosh^{-1}\left(\frac{x}{x-1}\right) \quad ; \quad f(x) = \tanh^{-1}(3x+1)$$

Exercise 12: Let f be a function defined by $f(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

1. Can we extend f by continuity at $x_0 = 1$
2. Calculate the derivative of f in $[0,1[\cup]1, +\infty[$
3. Prove that $\forall x \in]0,1[$ we have $\frac{2x}{1+x^2} < \tan^{-1}\left(\frac{2x}{1-x^2}\right) < 2x$
4. Deduce the limit $\lim_{x \rightarrow 0^+} \frac{1}{x} \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

Exercise 13: Let f be a function defined by $f(x) = (x^2 - 1) \ln \left| \frac{1+x}{1-x} \right|$.

1. Determine the domain of definition of f .
2. Calculate the limited development of f of order 3 in a neighborhood of 0.
3. Give the equation of the tangent at the point of abscissa 0, then specify its position to the graph of f .

Exercise 14: Let $f(x) = \ln(x + \sqrt{x^2 - 1})$ a function.

1. Determine D_f the domain of definition of f .
2. Study the continuity of f on D_f .
3. Study the derivability of f on $]1, +\infty[$, and calculate $f'(x)$.
4. Show that f admit a reciprocal function noted f^{-1} , with specifying the domain of definition and domain of arrival.
5. We set $x_0 = 2$ and $y_0 = f(x_0) = \ln(2 + \sqrt{3})$, calculate $f^{-1}(y_0)$.
6. Calculate the limit $\lim_{n \rightarrow 1^+} \frac{\ln(x + \sqrt{x^2 - 1})}{x-1}$ using the theorem of mean value.